

AOS104 RTG

Object Detection Systems

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1 About these operating instructions

Please read this chapter carefully before you begin working with this documentation and the AOS104 RTG.

1.1 Purpose of this document

These supplementary operating instructions are intended to give **technical personnel** of the machine manufacturer or operator instructions on the safe mounting, configuration, electrical installation, commissioning, operation, and maintenance of the AOS104 RTG object detection system.

Note **The supplementary operating instructions for the AOS104 RTG object detection system are only valid in conjunction with the underlying operating instructions for the AOS LiDAR (8016731).**

Unless otherwise specified in this document, the information in the underlying operating instructions for the AOS LiDAR will apply.

1.2 Target group

These supplementary operating instructions are intended for persons who integrate the AOS104 RTG object detection system into a crane system, and initially commission and operate it there. They are also intended for the planners, developers, and operators of crane systems for the handling of goods and containers.

1.3 Information depth

These supplementary operating instructions contain information about the AOS104 RTG system on the following topics:

- Product description
- Project planning
- Mounting
- Electrical installation
- Commissioning

When planning and using the AOS104 RTG system, technical skills are also required that are not covered by this document.

The applicable official and legal regulations at the application site must always be complied with when operating the system.

Note Further information about the device components used in the AOS104 RTG can be found in the accompanying operating instructions.

For carrying out the configuration and diagnostics of the AOS104 RTG, you will need the **SOPAS** SICK Engineering Tool (configuration software for the laser scanners), version V 3.0 or higher, and the **FSD Flexi Soft Designer** (configuration and diagnostic software for the control), version 1.7 or higher. Both of these tools are included in the scope of delivery.

1.4 Abbreviations used

AOS	Advanced object detection system
FSD	Flexi Soft Designer = software for Flexi Soft control for configuration and diagnostics
LMS	Laser scanner from SICK AG
RTG	Rubber tyred gantry
SOPAS	SICK Engineering Tool = software for laser scanner configuration and diagnostics
PLC	Programmable logic controller

1.5 Symbols used

Recommendation	Recommendations are designed to assist you in the decision-making process with respect to the use of a certain function or technical measure.
Note	Notes provide information about the features of a device, application tips, or other useful information.
1. / 2. ... Step by step	Instructions that must be carried out in the described order are referred to as step-by-step instructions and are indicated by numbered lists. Carefully read and follow the instructions for action.
➤ Action	Instructions for taking action are indicated by an arrow. Carefully read and follow the instructions for action.

2 Safety

This chapter concerns your own safety and the safety of the system operator.

- Please read through this chapter carefully before using the AOS104 RTG or the crane system monitored by the AOS104 RTG.

Note Please also observe all safety notes contained in the operating instructions for the components used.

2.1 Qualified personnel

The AOS104 RTG must only be mounted, commissioned, and serviced by adequately qualified safety personnel.

A qualified person

- has sufficient skills in the field of the respective equipment based on their technical training and experience **and**
- has been instructed by the manufacturer in system operation and all applicable safety guidelines **and**
- is familiar with all relevant country-specific occupational safety regulations, work safety regulations, guidelines, and generally accepted technical rules and standards (e.g., DIN standards, VDE regulations, country-specific rules) to such an extent that he/she is able to evaluate the safe condition of the power-driven machinery **and** he/she
- has access to and has read the operating instructions.

2.2 Applications of the system

The AOS104 RTG object detection system is a collision protection system on gantry cranes. The system is prepared for use on all RTGs and similar gantry cranes, such as RMG or ASC for the handling of goods and containers.

The system solution provides protection against collisions in the direction of travel (pathway), as well as optionally in cross travel with automated switching.

The ambient conditions typical in ports, such as rain, snow, or fog, are mastered by the AOS104 RTG system.

2.3 Intended use

The AOS104 RTG may only be used as described in section [2.2 Applications of the system](#). It may only be used by qualified personnel in the environment in which it was mounted and initially commissioned by qualified safety personnel in accordance with these operating instructions.

Note **The AOS104 RTG system is not a safety device for human protection and it therefore does not comply with any safety standards. For safety applications, please contact SICK AG.**

2.4 General safety notes and protective measures



NOTE

These supplementary operating instructions must be made available to the operator of the crane system on which the AOS104 RTG system is used, along with the basic AOS LiDAR underlying operating instructions.

The operator must be instructed by qualified safety personnel and read the operating instructions.

3 Product description

This chapter provides information on the special features of the AOS104 RTG object detection system. It describes the construction and operating principle of the system, in particular the interaction of the different components.

Note Always read this chapter before you mount, install, and commission the system.

3.1 Scope of delivery

The AOS104 RTG consists of four pre-configured LMS111 laser scanners and the Flexi Soft control, which comprises the main module and two I/O expansion modules. The system is supplied with the required accessories.



Fig. 1: Main components of the AOS104 RTG

Thorough check for completeness

Important note

- It is recommended that you carefully check for and report transport damage of any kind as soon as possible after receiving the system.
- Also verify that the delivery includes all components listed on the delivery note.

Part No.	Quantity	Description
1064544	1	AOS104 RTG
Consisting of:		
Laser scanner		
1047516	4	Laser scanner LMS111-10100S01
2062346	4	Connection box for power, I/O, and RS-232/RS-422 data (not Ethernet), with three pre-wired M12 cables
2046458	4	Weather hood 270°
2046025	4	Mounting bracket
6030928	1	Ethernet connection cable SSL-2J04-G10ME
Control		
1043783	1	Flexi Soft main module FX3-CPU000000
1043700	1	Flexi Soft system plug FX3-MPL000001
1044125	2	Flexi Soft I/O module FX3-XTIO84002
6034574	1	Configuration cable DSL-8U04G02M025KM1 for connecting the configuration connection of the Flexi Soft control to the USB interface of the configuration PC
2073337	1	USB stick with configuration file (prepared configuration for control and laser scanner) as well as device and system documentation

Tab. 1: AOS104 RTG scope of delivery

Note The “SOPAS” and Flexi Soft Designer configuration tools can be downloaded at www.sick.com.

3.2 System components

All components of the AOS104 RTG system are prepared for their use on crane systems for the handling of goods and containers.

LMS111 laser scanner with weather hood

The LMS111 laser scanners are designed for outdoor use in harsh ambient conditions and are fitted with a weather hood which is part of the scope of delivery. All input and output signals from the laser scanners are directly connected to the central Flexi Soft control.



Fig. 2: LMS111 laser scanner with weather hood

The monitoring radius of the LMS111 laser scanners is 20 m/18 m for black objects with 10% remission. The scanning angle is 270°.

The laser scanners feature automated contamination measuring for the front screen. The laser scanners are also equipped with an internal heater.

The Flexi Soft control

The Flexi Soft control for the AOS104 RTG system consists of the **CPU0** central main module with a configuration memory and two **XTIO** I/O expansion modules.

The two I/O expansion modules supply the input and output signals from the AOS104 RTG. The communication with the connected laser scanners and with the higher-level crane control occurs via the switching signals.

One XTIO I/O expansion module is used for every two laser scanners.

The test target

The self-diagnosis of the object detection system occurs via the continuous detection of an external test target. During this process, the corresponding scanner monitoring field is infringed by the test target. Only by using external test targets can the full functional range of the AOS104 RTG system be ensured.

At least one test target is required to continue the self-diagnosis of the system. It is also possible to use part of the crane as a test target.

Configuration

The configuration of the Flexi Soft control contains all the functions required to evaluate the system based on the motion scenarios and to cyclically test the laser scanners.

It is provided with the supplied USB stick and is to be loaded into the Flexi Soft control.

The configuration of the laser scanner is prepared for use on the gantry cranes and is already pre-set in the devices (factory settings). You only have to edit the field dimensions and shape of the monitoring fields according to the actual demands and installation positions of the laser scanners.

3.3 System functionality

The AOS104 RTG provides protection against collisions by means of four laser scanners, which are mounted onto each of the four wheel bogies on the gantry crane.

The laser scanners monitor the direction of travel for longitudinal and, optionally, cross travel.

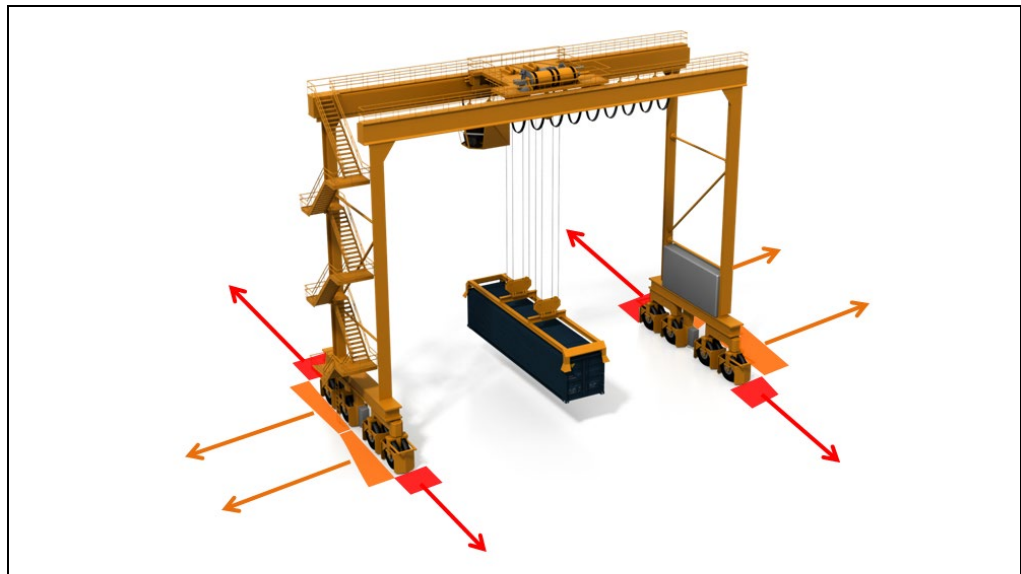


Fig. 3: System functionality

If an object is detected in the monitored area, the system sends a slow-down signal (warning signal) to the crane, followed by a stop signal, in case the gantry crane is to continue traveling.

For this purpose, the output signals of the laser scanners are evaluated and processed by the Flexi Soft control.

4 Project planning

4.1 Monitoring fields and evaluation cases

Monitoring fields

In each laser scanner, the following standard monitoring fields are available:

Field	Monitoring scenario
Stopping field 1	Monitoring of the forward/backward direction of travel (pathway).
Warning field	Monitoring field for speed reduction in the direction of travel.
Stopping field 2	Monitoring of the cross travel.
Test field	Monitoring field for detecting the external test target.
Field 5	Field for free use

Tab. 2: Pre-configured monitoring fields

Note

All fields have predefined dimensions and have to be edited according to the operational requirements and the mounting of the laser scanners. Field 5 is empty as an exception and must be defined for use.

Evaluation cases

Each monitoring field is linked to an evaluation case (EVC). An evaluation case determines the way the monitoring field is evaluated and which switching output is addressed. Evaluation cases can be activated via input signals depending on the required scenarios. Ten different evaluation cases can be configured and permanently saved in each laser scanner. Depending on the input signals that are present for the evaluation case, the laser scanner switches between the cases.

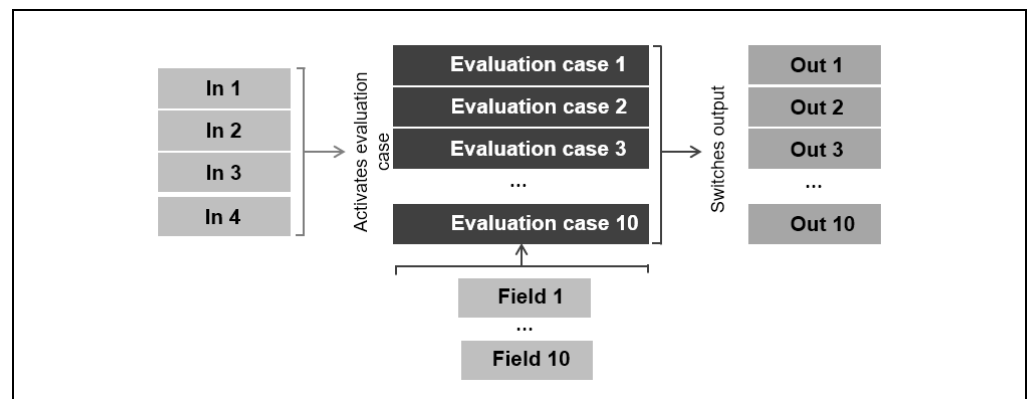


Fig. 4: Activating an evaluation case

4 PROJECT PLANNING

4.1.1 Pre-configured evaluation cases (EVCs)

The AOS104 RTG is supplied with 10 evaluation cases. It is not necessary to use all evaluation cases.

Evaluation case	Input 1 (signal from the Flexi Soft)	Input 2 (signal from the crane control)	Output	
1	0	0	1 (stop 1)	Forward/backward (pathway)
2	0	0	2 (test)	
3	1	0	2 (stop 1)	
4	1	0	1 (test)	
5	0	1	1 (stop 2)	Cross travel
6	0	1	2 (test)	
7	1	1	2 (stop 2)	
8	1	1	1 (test)	
9	X	0	3	Warning field
10	X	1	3	Optional field

Tab. 3: Pre-configured evaluation cases and switching signals

Legend Prepared LMS111 evaluation cases:

1 = 24 V DC

0 = 0 V

X = input not relevant

4.1.2 Input and output signals

All input and output signals from the laser scanners are directly connected to the Flexi Soft control.

The Flexi Soft control analyzes the status signals from the laser scanners based on the stored monitoring scenarios and transmits corresponding **stop** and **warning signals** to the higher-level crane control.

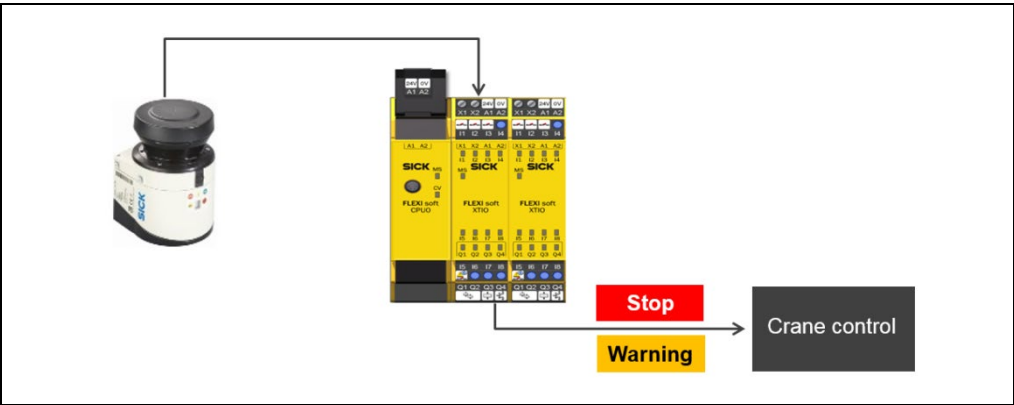


Fig. 5: Input and output signals

Cross travel switching signals

The crane control signals a change to cross travel via a separate input signal to the Flexi Soft control (**I4** input of the **XTIO 1** expansion module). The crane control signal is transferred to the laser scanner **In2** input via the **Q3** output. It activates the evaluation cases for the **Stop2** monitoring field.

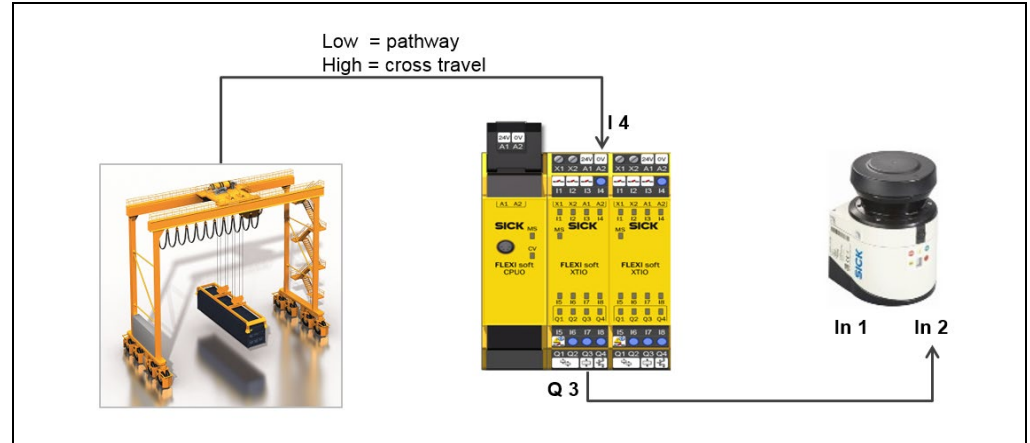


Fig. 6: Cross travel switching signals

Self-test switching signals

The Flexi Soft control also handles the internal self-testing of the laser scanners. To do this, the Flexi Soft control **cyclically** sends a test command signal to the **In 1** input of all laser scanners.



You can find additional information on the system self-test in the AOS LiDAR operating instructions.

The change in assignments causes the switching outputs to respond automatically with a signal change from High to Low or, conversely, from Low to High.

Note

The test cycles run in the background of the evaluation and do not influence the system reaction during regular object detection

4.2 Signaling of the stop and warning signals

Different versions of gantry cranes require individual signaling or management of the stop and warning signals.

There are two types of stop signals:

- Individual stop signals of the wheel bogie (corner signals)
- Direction-dependent stop signals

4.2.1 Individual stop signals per wheel bogie in pathway travel (corner signals)

When retrofitting an existing crane, the crane control might expect individual stop signals from the wheel bogies frames.

In this case, the AOS104 RTG can be set up in such a way that the stop signals are output as so-called **corner signals**. The signals are output from both I/O expansion modules of the Flexi Soft control via the **Q1** and **Q2** outputs.

Then the crane control does not have to be modified. It can process the signals that it recognizes. A corresponding configuration is prepared for the Flexi Soft control. The configuration of the laser scanners does not change as a result of this.

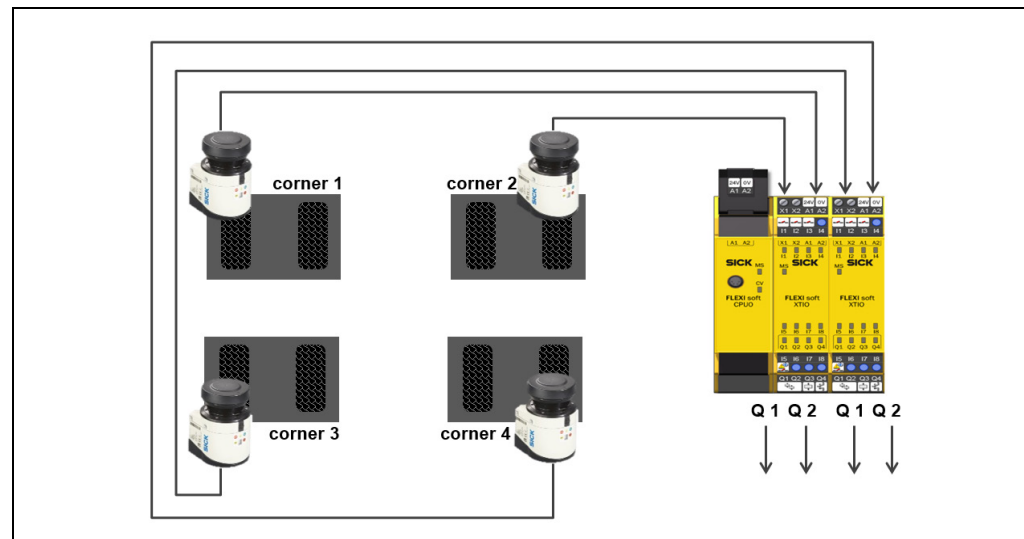


Fig. 7: Stop signals per wheel bogie (corner signals)

4.2.2 Direction-dependent stop signals in pathway travel

Alternatively to the corner signals, the stop signals can be output subject to the direction of crane movement. The stop signals of two respective laser scanners are combined in the Flexi Soft control to form one stop signal.

The higher-level crane control signals the direction of travel via a static signal on the **I4** input (module **2**) of the Flexi Soft control.

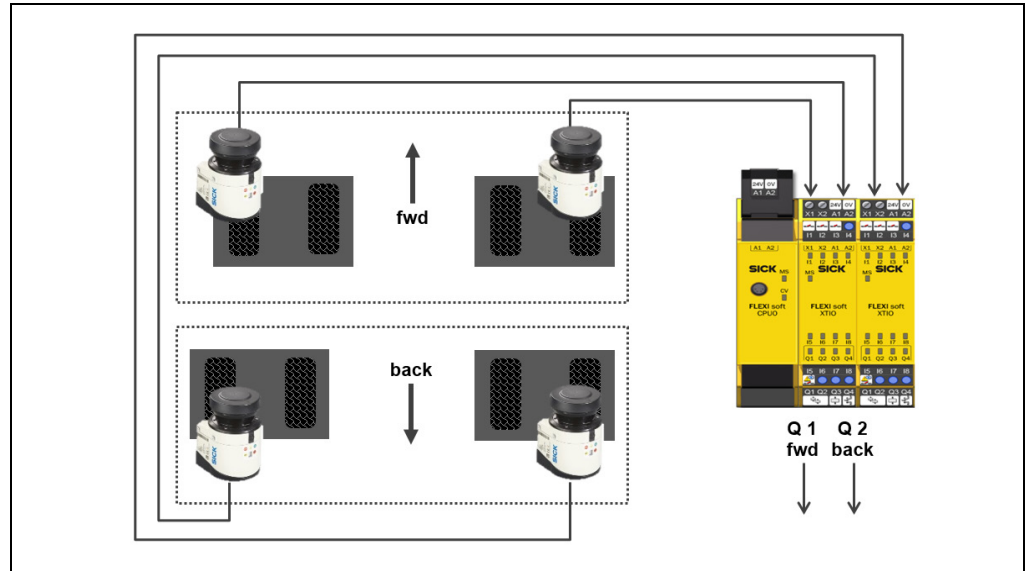


Fig. 8: Direction-dependent stop signals

Note The direction-dependent stop signals are output on the **XTIO 1** I/O expansion module to the **Q1** (forward) and **Q2** (backward) outputs. In cross travel, direction-dependent stop signals cannot be output. The output is generally done to the **fwd** or **back** outputs.

4.2.3 Reducing speed in pathway travel via warning fields

AOS104 RTG provides warning fields for pathway travel which are upstream to the actual stopping field. The warning fields can be used by gantry cranes without a brake ramp to reduce speed.

If a warning field is infringed, the speed of the crane can be reduced via the crane control before a final stop signal can be sent.

4.2.4 Stop signals in cross travel

The route of cross travel is monitored by a separate monitoring field.

Due to the fact that crane drivers and supporting safety personnel, if present, usually pay greater attention during this process, as well as the reduced crane speed, a warning field is not provided.

In cross travel, all monitoring fields work simultaneously and are output individually from the Flexi Soft control regardless of the direction of travel.

Note The monitoring case of the cross travel is dependent on the physical positioning of the laser scanners based on the mounting (the visual range of the laser scanners can be restricted).

4.3 Dimensioning the monitoring fields

To ensure that the AOS104 RTG functions correctly and to avoid unintended shutdowns, the following rule applies due to the size of the monitoring fields:

As large as necessary. As small as possible.

The following figure shows the principle location and alignment of the fields:

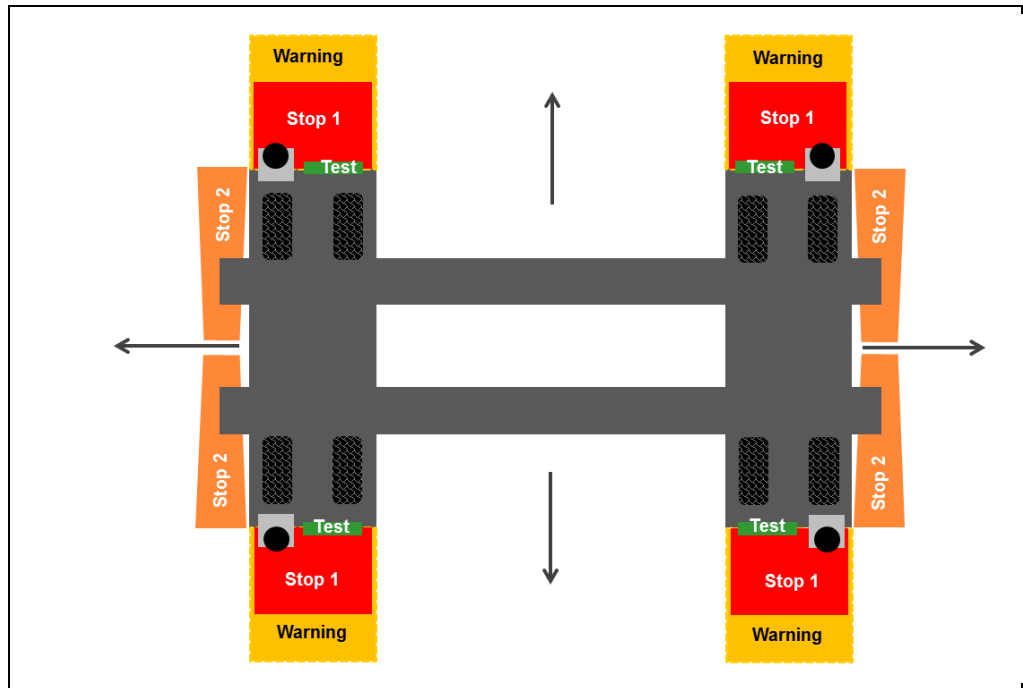


Fig. 9: Principle location of the monitoring fields

Size of the stopping fields

The size of the stopping field in the direction of travel (forward/backward) depends on the crane's stopping distance. The stopping distance S_A is calculated as follows:

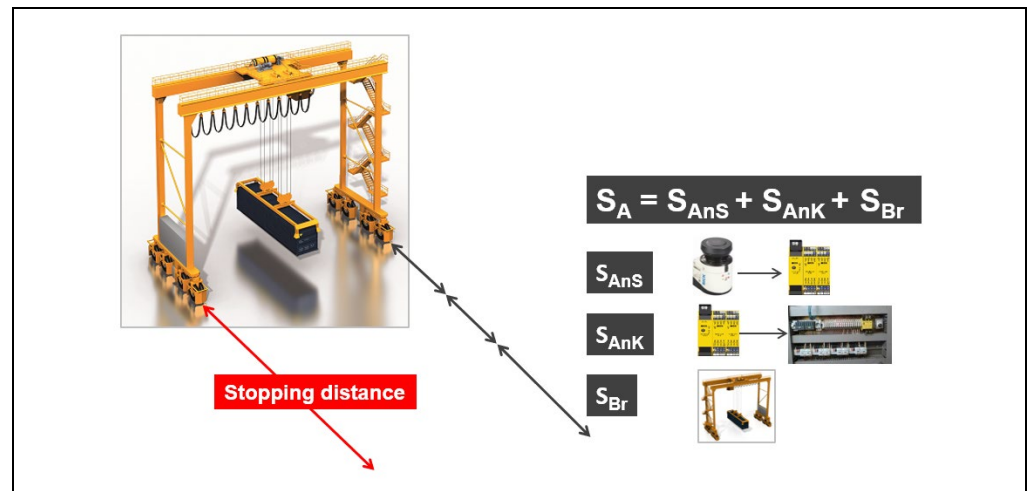


Fig. 10: Calculating the stopping distance

$S_A = S_{AnS} + S_{AnK} + S_{Br}$	
Term	Meaning
S_{AnS}	Distance covered during the response time of the AOS104 RTG.
S_{AnK}	Distance covered during the response time of the crane control.
S_{Br}	Gantry crane braking distance according to documentation concerning full load and maximum speed.

If required, individual values can be ascertained in the field using representative stop procedures.

In addition, a 10% allowance is recommended when calculating the size of the stopping field to compensate for a reduction in brake function due to wear and tear, for example.

- Note** ► After commissioning, always implement several stop procedures to verify the size of the field (see chapter [7.2 Checking the stopping and warning response of the system](#)).

Size of the warning fields

The size of the warning fields is generally based on the response time of the crane or the runout distance until the speed is reduced.

Typically, several test drives are required to determine the field size. A feasible initial size for a warning field is:

The size of the stopping field in the direction of travel + 50%.

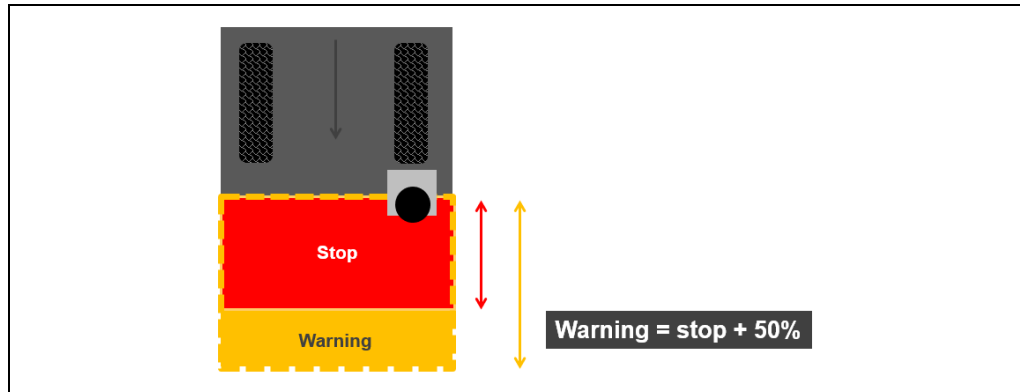


Fig. 11: Size of the warning fields

- Increase or decrease the output field during the test drives systematically until the slowing-down and stop procedures start to converge.
- Monitor the behavior of the crane during the test drives.

Width of the monitoring fields in pathway travel

The width of the monitoring fields depends on the width of the wheel bogie.

We recommend selecting the width so that superstructures on the side of the gantry crane such as steps or protective hoods are protected against collision.

The mechanical limit switches (so-called whisker switches) that are used for lane guidance between the containers can also be covered with the applicable geometry of the monitoring fields.



Fig. 12: Width of the monitoring fields in longitudinal travel

Width of the monitoring fields in cross travel

The route of cross travel can be monitored by separate monitoring fields. The alignment of the stopping fields for cross travel depends on where the laser scanner was mounted and whether it rotates with the wheel bogie or not.

Note Check in which direction the laser scanner will rotate with the wheel bogie.

- If the laser scanner rotates with the wheel bogie, you must align the stopping fields for cross travel in the direction of the stopping fields for longitudinal travel. The stopping field for longitudinal travel is deactivated for cross travel.

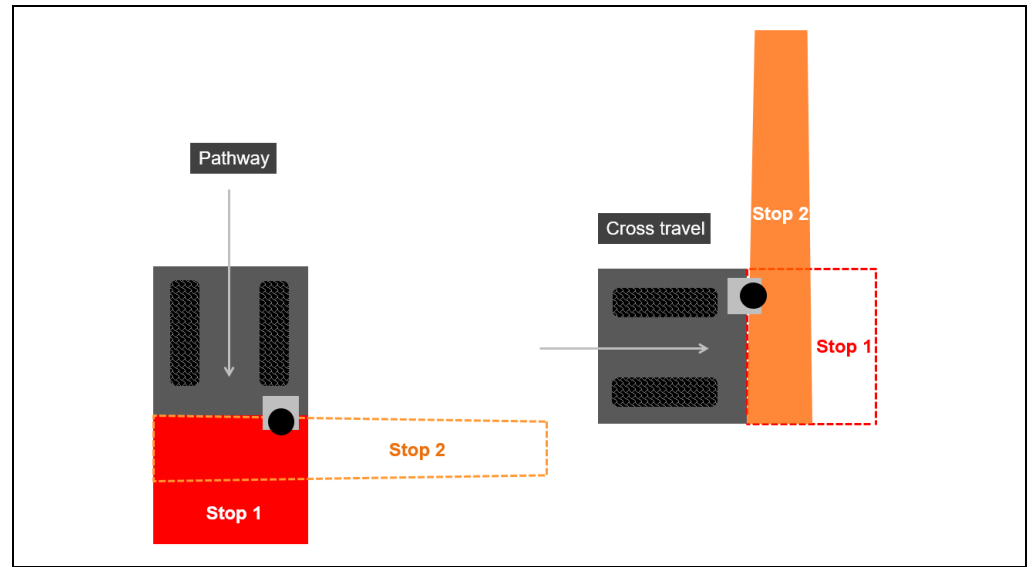


Fig. 13: Width of the monitoring fields in cross travel (LMS rotates with wheel bogie)

- If the laser scanner does not rotate with the wheel bogie, the stopping fields for cross travel are aligned sideways.

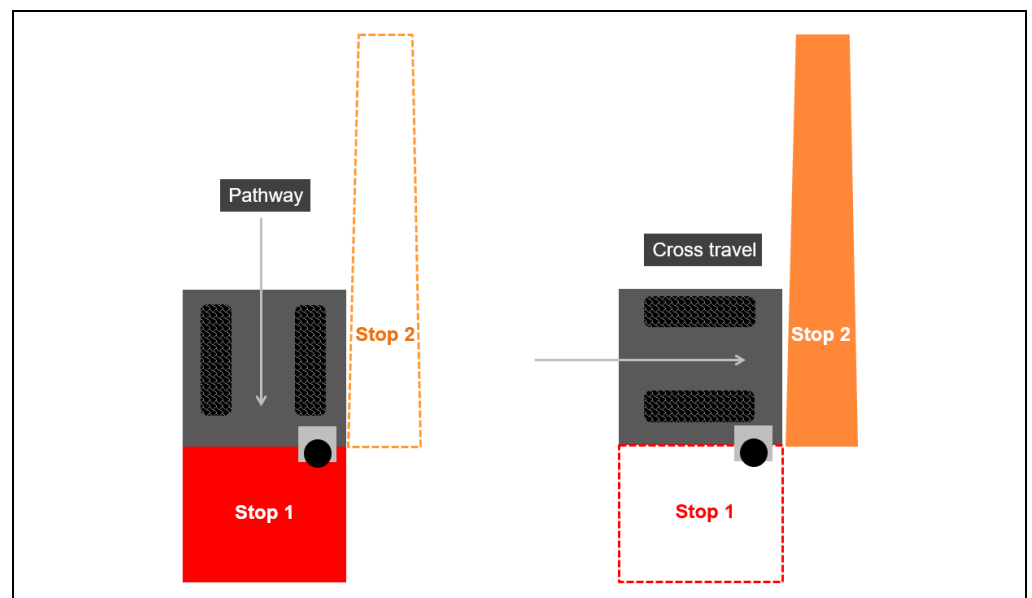


Fig. 14: Width of the monitoring fields in cross travel (LMS does not rotate with wheel bogie)

Note If the laser scanner turns with the wheel bogie inwards, the cross travel can only be monitored with an additional measure (e.g. with additional laser scanners on the outside).

Size and position of the test field

The system's self-monitoring test fields are typically aligned with a suitable flat object on the crane or directly with a flat area of the crane body.

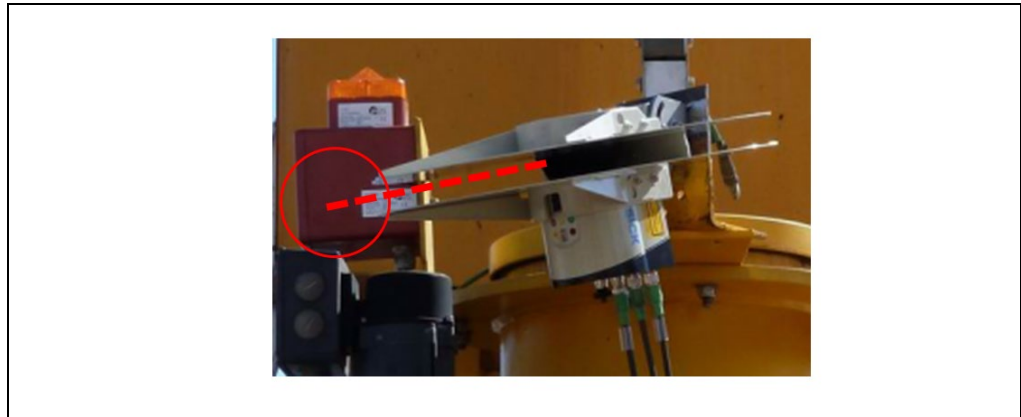


Fig. 15: Size and position of the test field

In order to avoid reflections on the surface of the external test target, the area of the test target must not be covered with a glossy coating.

Ideally, the test target is positioned at a distance of 600 mm from the laser scanner (± 100 mm). The width of the test field on the test target must be at least 100 mm.

The laser beam should hit the test target at a right angle. An unfavorable alignment at an acute angle is to be avoided.

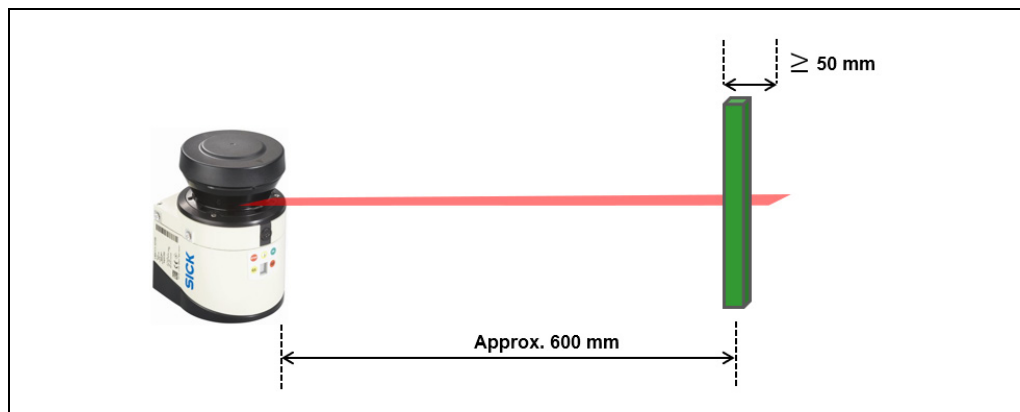


Fig. 16: Positioning test field on the test target (1)

Note Because the LMS111 laser scanner has a scanning angle of 270° , test targets can also be positioned to the rear or the side.

5 Mounting

5.1 Mounting the laser scanners

The aim of the AOS104 RTG is to avoid collisions with containers, vehicles, or other objects that are positioned on the lanes of the gantry crane.

The reliable and problem-free operation of the system depends primarily on the proper mounting and alignment of the laser scanners.

The laser scanners are mounted on the wheel bogie. They monitor the width of the wheel bogie including the superstructures on the side of the crane.

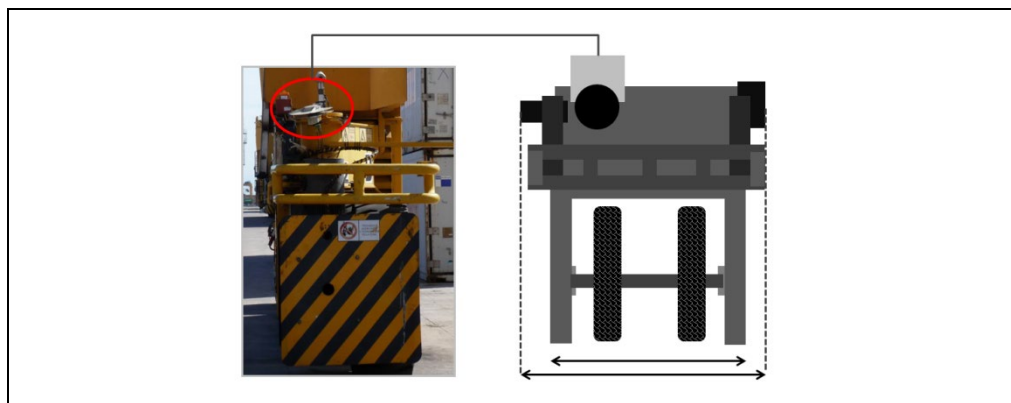


Fig. 17: Mounting the laser scanners – general

5.1.1 Protected mounting location

Depending on the type of gantry crane, the laser scanners can be integrated into the chassis of the crane or mounted to another protected position in the wheel bogie.

► Please ensure that the mounting location provides sufficient **mechanical protection**.



Fig. 18: Mounting the laser scanners – protected mounting location



WARNING

The laser scanners must not protrude unprotected beyond the wheel bogie.



5.1.2 Alignment

The laser scanners must be mounted and aligned in such a manner that they can detect objects that penetrate the monitored area. Only with **optimal positioning** and **alignment** can the AOS104 RTG system provide adequate protection against collisions.

- ▶ Always be sure that the laser scanner has an unobstructed view of the areas that are to be monitored in all monitoring scenarios. The mounting position must ensure an optimal sight range for pathway travel and - provided the mounting or crane construction makes this possible - for cross travel as well.
- If the laser scanner rotates with the wheel bogie, a clear view is ensured during rotation in longitudinal and cross travel.

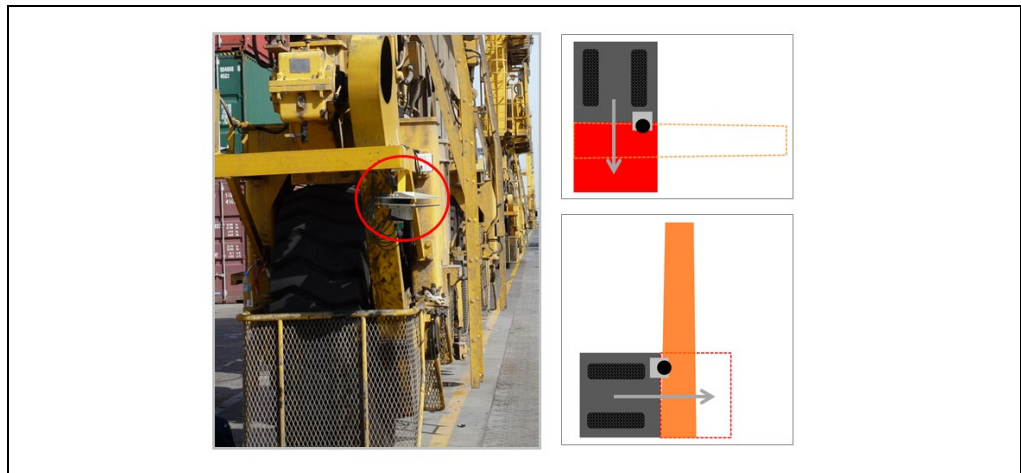


Fig. 19: Mounting the laser scanners – alignment (LMS rotates with wheel bogie)

- If the laser scanner does **not** rotate with the wheel bogie, the mounting position must ensure that the monitored area is also covered in cross travel.

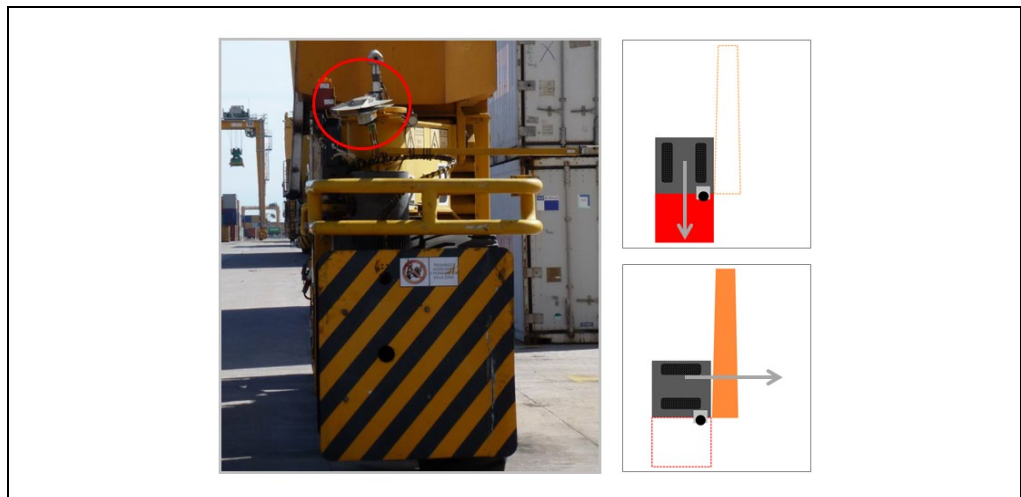


Fig. 20: Mounting the laser scanners – alignment (LMS does not rotate with wheel bogie)

- Due to the possibly higher mounting position of a laser scanner, make sure the inclination angle of the scan plane is suitable.

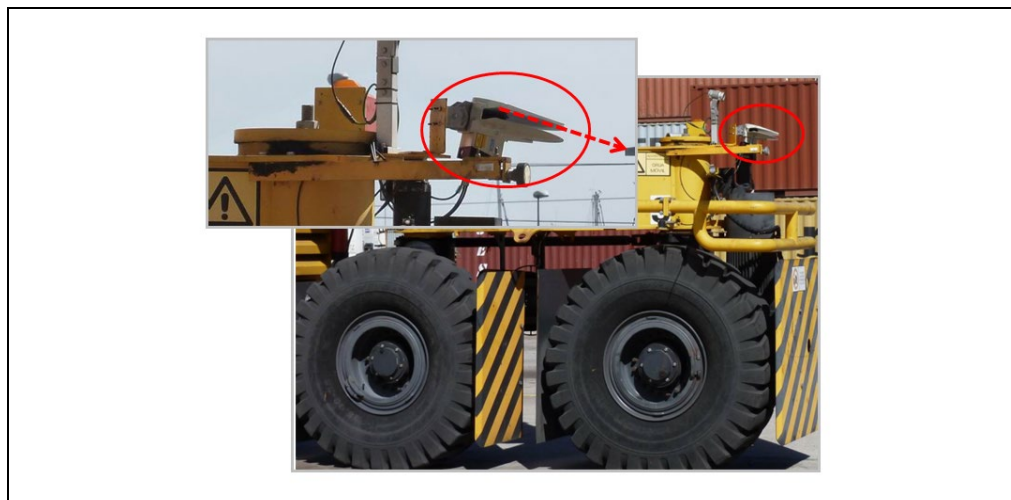


Fig. 21: Mounting the laser scanners – tilt angle

Tip Mount the LMS111 with weather hood a little to the side. This improves monitoring of the sides.

5.1.3 Mounting height and tilt angle

In most cases, the mounting height of the laser scanner arises from the existing mechanical conditions on the wheel bogie. Construction conditions mean that it is often only possible to mount it above the wheel. The inclination angle must be set depending on the mounting height.

- As the monitoring fields are mostly tilted downwards, care must be taken to ensure that the field boundaries that are separated by a distance of several meters do not impact the route due to rocking movements by the crane and trigger a stop erroneously.

As the following figure illustrates, the distance **d** between the end of the monitoring field (see arrow) and the ground should be observed.

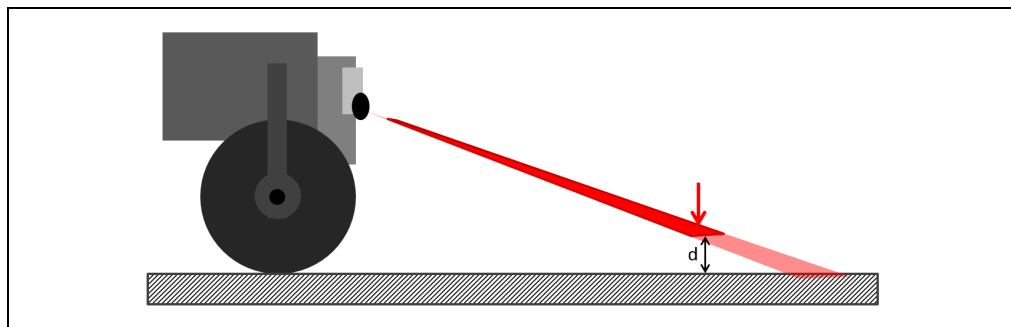


Fig. 22: Mounting the laser scanners – mounting height and tilt angle

As a practical value for gantry cranes, the distance can be ascertained as follows:

$$d = \text{approx. } 1 \text{ m at field sizes of } 10 \text{ to } 12 \text{ m}$$

If the distance **d** is too small, the monitoring field will rock with the crane on the ground and trigger a stop erroneously.

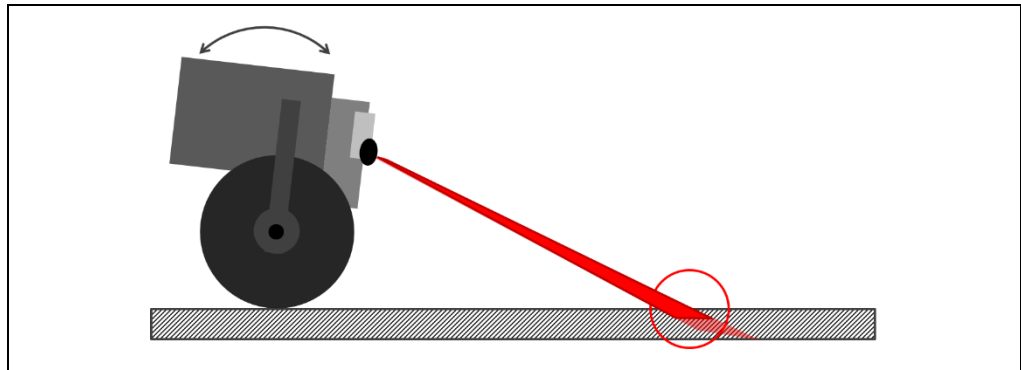


Fig. 23: Mounting the laser scanners – monitoring field rocks on the ground

Note for detection of small objects and on the inclination angle

Depending on the assembly height of the laser scanner and the resulting inclination angle, objects located directly in front of the wheel bogie of a **stationary** or **approaching** crane may not be detected in some circumstances.

The assembly height and the inclination angle should thereby always take into account the required brake path and the detectability of small objects in the route. Depending on the mounting position, it may be necessary to make a compromise.

Mounting height and inclination angle are set correctly in the following examples: If an object infringes the monitoring field when a crane is moving, the crane is stopped.

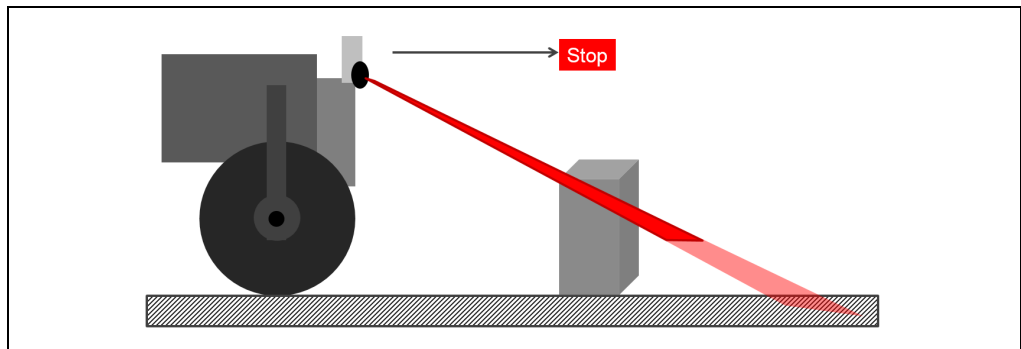


Fig. 24: Mounting the laser scanners – stop in the event of a field violation

If a small object is located immediately in front of the wheel bogie when the gantry crane is restarted, it might not be detected.

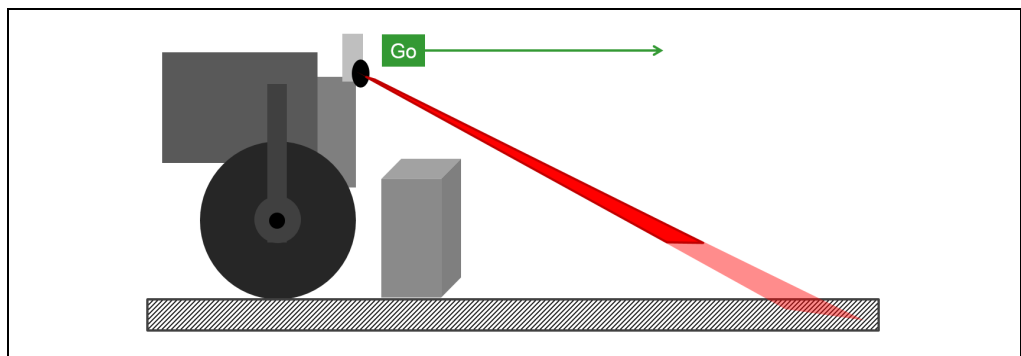


Fig. 25: Mounting the laser scanners – no stop signal if small objects are present

5.1.4 Checking if the alignment of the scan plane is correct

The scan plane can be correctly aligned using the scan finder, which you can order as a SICK accessory.

Device	Accessories	Description	Part number
	Alignment aid	Alignment aid for detection of infrared light from SICK sensors.	2101720

Alternatively, you can place objects of suitable heights in the scan plane and check it in the live monitor in the SOPAS configuration software.

Note After aligning the laser scanners, test drives are recommended to check that the entire system is functioning correctly (see chapter [7.2 Checking the stopping and warning response of the system](#)).

5.2 Mounting the test target

The test target must be mounted in such a way that the laser beam impinges on the surface of the test target ideally perpendicularly.

It must always be located **within the scanning range** of the respective laser scanner, but it must be located outside of the monitoring field.

Check whether there is a system part that would be suitable for use as a test target. If none can be found, mount a suitable test target.

5.3 Mounting the Flexi Soft control

The Flexi Soft control is situated in a suitable location in the control cabinet of the crane control.

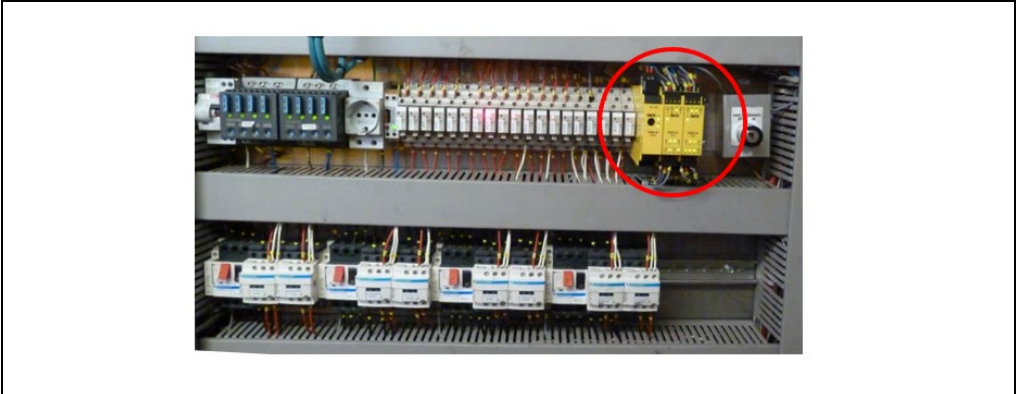


Fig. 26: Mounting the control

Note

- In order to accommodate the leads and wiring of the signals, there must be enough free terminals available in the control cabinet (see [6 Electrical installation](#)).
- Avoid assembly positions near high-voltage components or contactors (EMC interference).

6 Electrical installation



HAZARD

Disconnect the power to the system

- ▶ Make sure that all the components of the system are disconnected from the power supply during electrical installation work.



HAZARD

Risk of injury due to electrical current

- ▶ Standard safety requirements must be met when working on electrical systems.
- ▶ The power supply must be disconnected when attaching and detaching electrical connections.

6.1 Electrical connection

The electrical connection between the Flexi Soft control and the individual laser scanners is established by means of:

- Already-existing and suitable cables or
- Individual installation cables from the Flexi Soft control to the local connection box on every laser scanner or
- Common installation cables with optional distributor boxes

Note ▶ **Shielded** cables must always be used.

6.1.1 Connection by means of individual installation cables

Shielded cables with at least 7 wires are required. A cable cross-section of 1.5 mm² is sufficient.

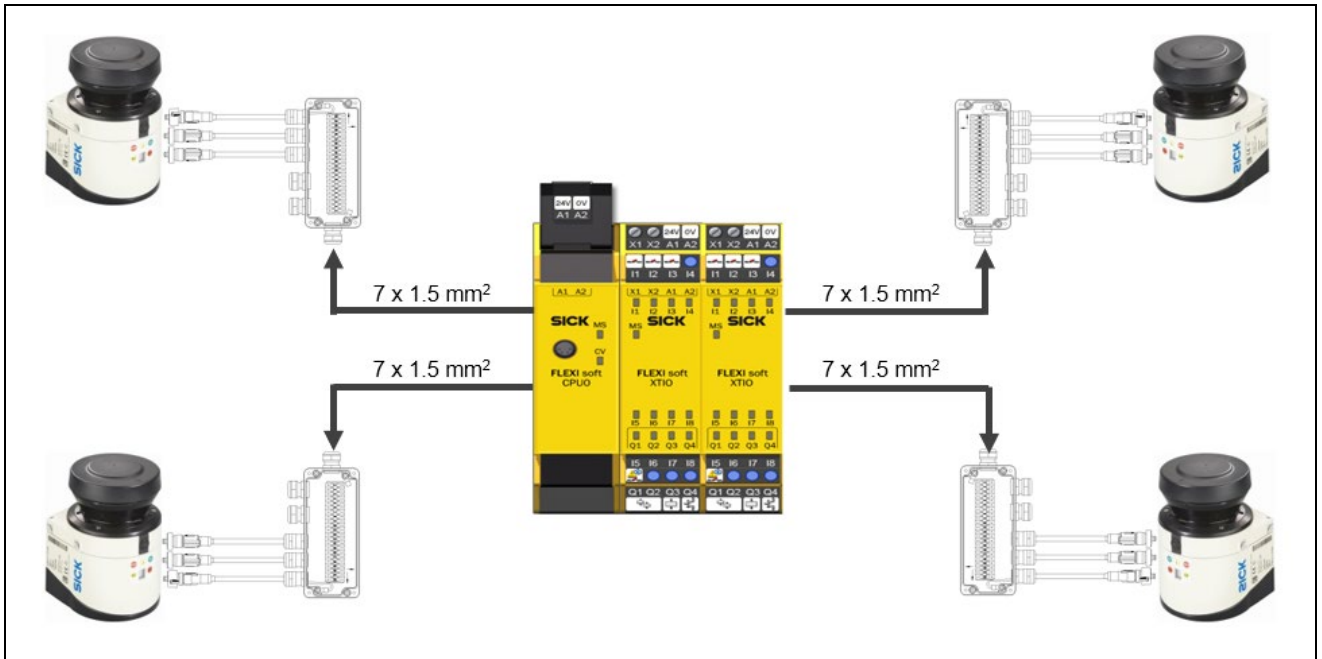


Fig. 27: Electrical installation by means of individual installation cables

Mounting the connection box

The terminal box is mounted separately either next to, above or below the laser scanner.

Connecting wires in the connection box

The connection box contains a terminal strip with screw terminals.

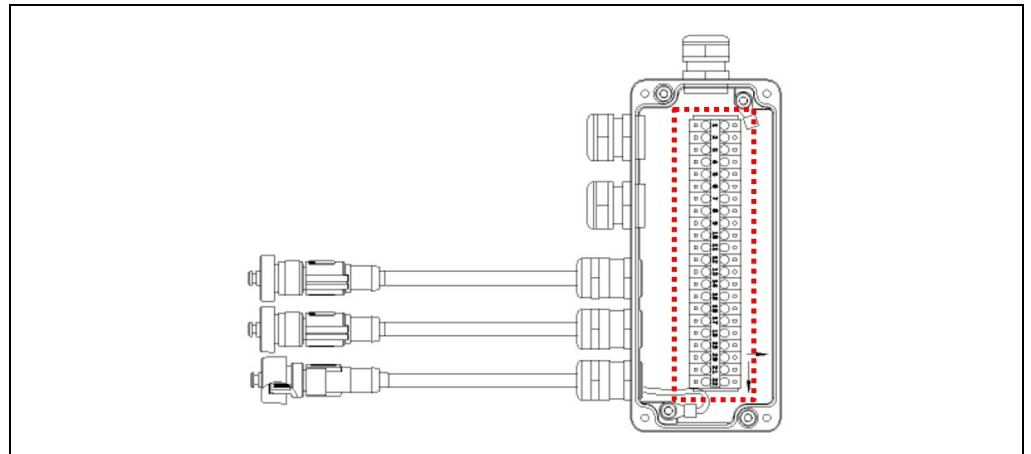


Fig. 28: Local connection box on the LMS with screw terminals

► Refer to the figure below for the pin assignment in the connection box terminal block.

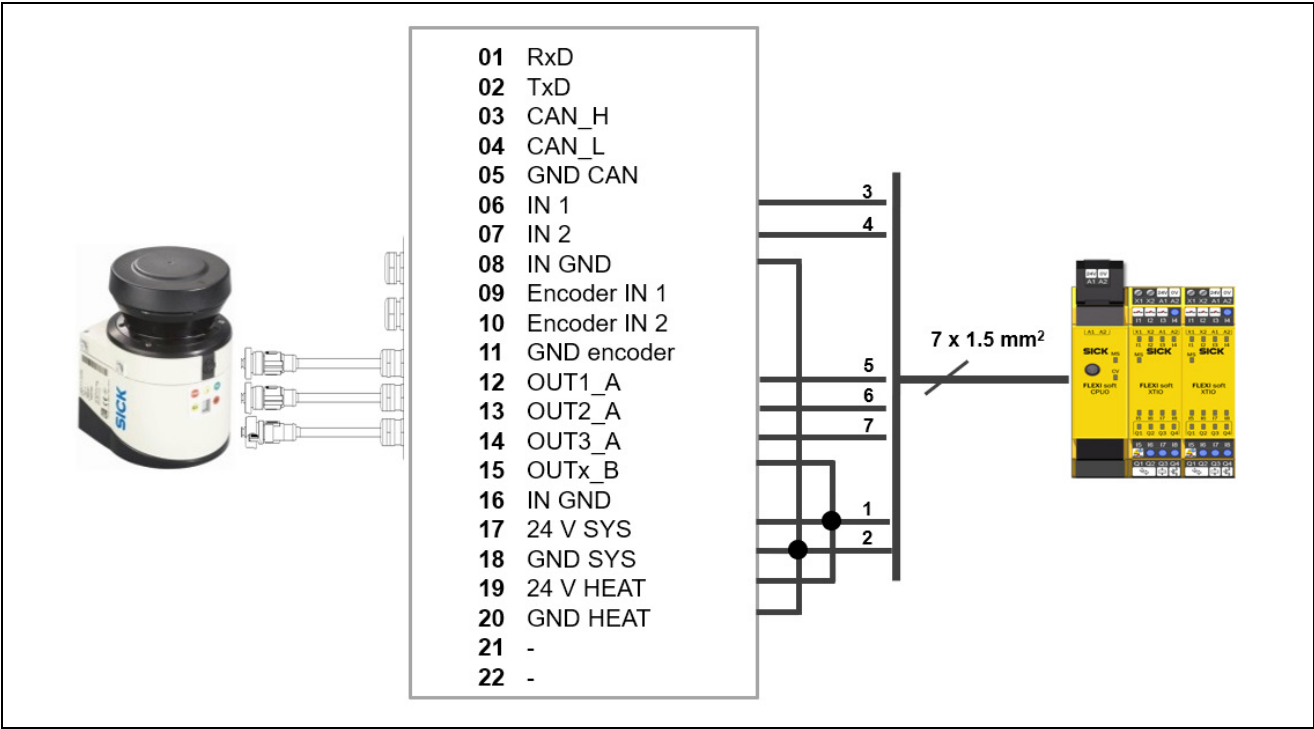


Fig. 29: Pin assignment in the local terminal box

- Notes**
- The jumpers between terminals 08, 18, and 20, and between terminals 15, 17, and 19 are to be placed on-site.
 - If **two separate power supply units** are used for the sensor and the heater, terminals 17 to 19 and 18 to 20 are **not** bridged. The 0-V outputs of both power supply units must be connected for potential equalization.

Connecting the laser scanner to the local connection box

The laser scanners are connected to the terminal box using pre-wired M12 plug connectors. The connections cannot get mixed up.

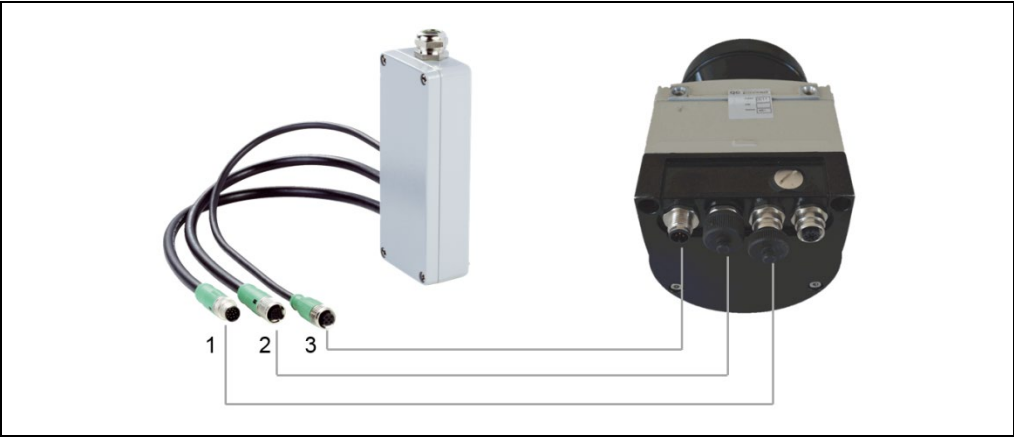


Fig. 30: Connecting the laser scanner to the connection box

Cable no. (in picture)	Laser scanner connection
1	I/O cable
2	Data/Input cable
3	Voltage supply

Tab. 4: Connecting the laser scanner to the connection box

6.1.2 Connection by means of common installation cables and distributor box

The shielded cable to the distributor box must have at least 12 wires. A cable cross-section of 1.5 mm^2 is sufficient.

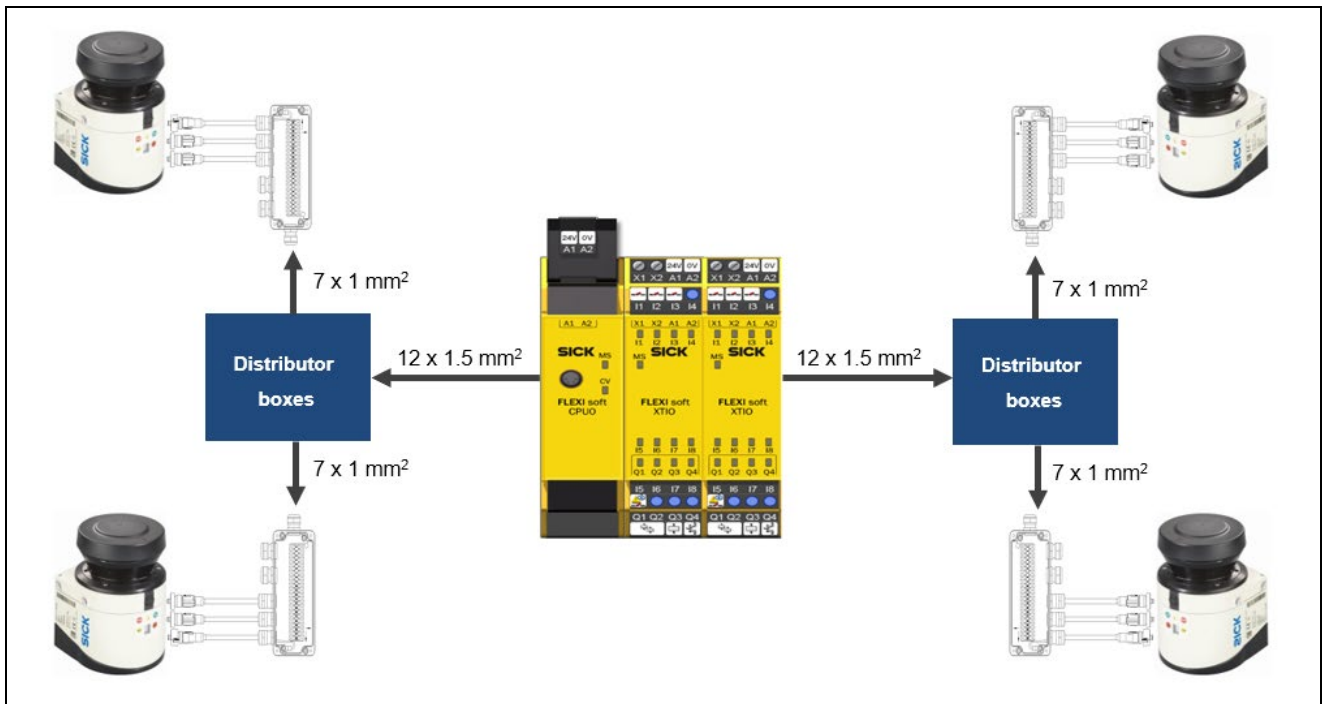


Fig. 31: Connection by means of a distributor box

Note The **voltage supply**, **ground** and **control input** of the respective laser scanners are fed as **one signal**.

Connecting and wiring distributor boxes

- Refer to the figure below for a possible pin assignment in the distributor box (suggestion).

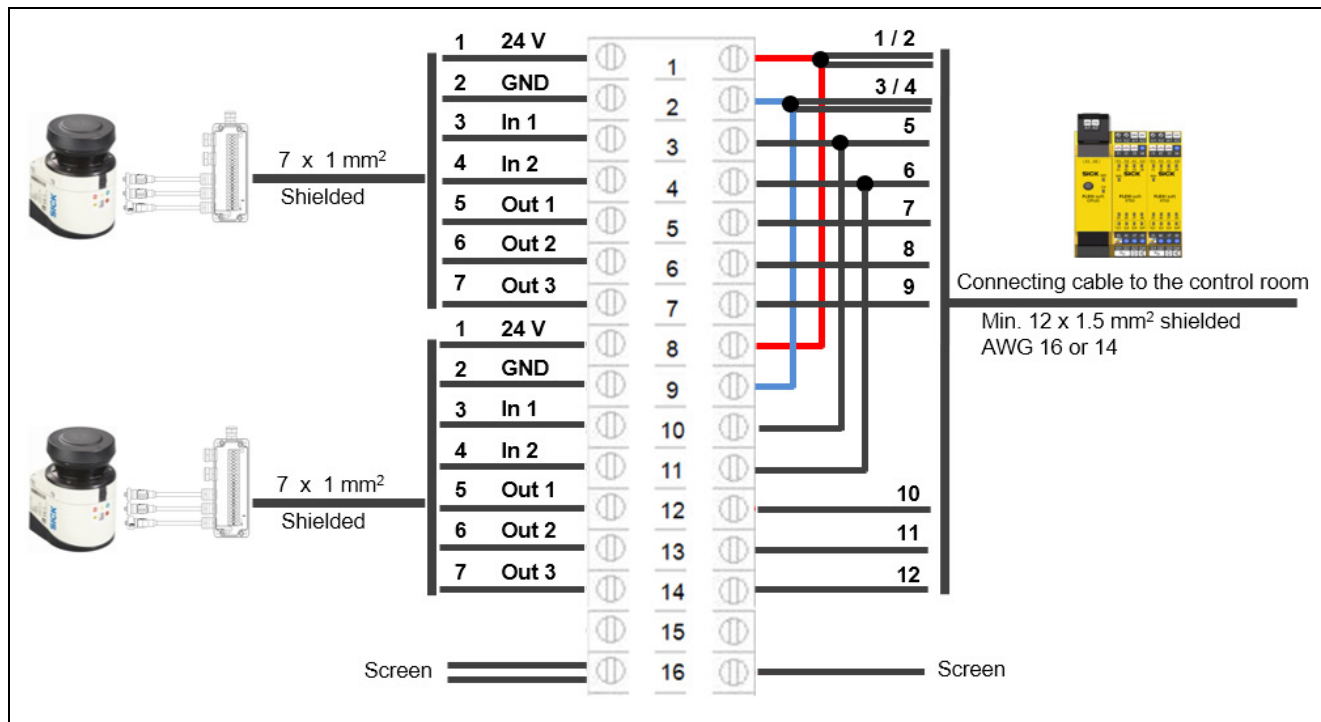


Fig. 32: Electrical installation by means of a distributor box (suggestion)

- Notes**
- The jumpers between terminals 01 and 08, and between terminals 02 and 09, are to be placed on-site.

6.1.3 General notes about the cables

Wire cross-sections of 1 mm² are sufficient for the I/O signals. For the voltage supply, the power requirement including the laser scanner heater may need to be taken into account.

- Check the performance of the power supply unit.
- With long cables, also take the voltage drop that occurs when the heater is in operation into account.

Depending on the length of cable, wire cross-section of between 1 mm² and 2.5 mm² are sufficient here.

Note The Flexi Soft control and the laser scanners must be connected to the same voltage supply.

If using multiple voltage supplies, a 0 V potential equalization between the individual power supplies must be established.

6.2 Flexi Soft pin assignment

The terminal assignment on the switching outputs of the Flexi Soft control depend on the loaded Flexi Soft configuration. Different configurations require different pin assignments.

Pin assignment of the inputs

The pin assignment of the inputs is the same for all configurations.

Expansion module	Input (Flexi Soft)	Source	Note
XTIO 1	I1	LMS 1 OUT 1	
	I2	LMS 1 OUT 2	
	I3	LMS 1 OUT 3	
	I4	Crane cross travel	Not connected if not used.
	I5	LMS 2 OUT 1	
	I6	LMS 2 OUT 2	
	I7	LMS 2 OUT 3	
	I8	-	
XTIO 2	I1	LMS 3 OUT 1	
	I2	LMS 3 OUT 2	
	I3	LMS 3 OUT 3	
	I4	Direction	Not connected if not used.
	I5	LMS 4 OUT 1	
	I6	LMS 4 OUT 2	
	I7	LMS 4 OUT 3	
	I8	-	

Tab. 5: Pin assignment of the inputs on the Flexi Soft control

Pin assignment of the switching outputs in the “Corner signals” operating mode

Expansion module	Output (Flexi Soft)	Function (“Corner signals” operating mode)	Note
XTIO 1	Q1	Stop corner 1	
	Q2	Stop corner 2	
	Q3	To all LMS In2	For cross travel
	Q4	To all LMS In1	Test command
XTIO 2	Q1	Stop corner 3	
	Q2	Stop corner 4	
	Q3	Warning/Speed reduction	Common output
	Q4	Test error	

Tab. 6: Pin assignment of the switching outputs in the “Corner signals” operating mode

Pin assignment of the switching outputs in the “Direction-dependent signals” operating mode

Expansion module	Output (Flexi Soft)	Function (“Corner signals” operating mode)	Note
XTIO 1	Q1	Forward stop	
	Q2	Backward stop	
	Q3	To all LMS In2	For cross travel
	Q4	To all LMS In1	Test command
XTIO 2	Q1	-	
	Q2	-	
	Q3	Warning/Speed reduction	Common output
	Q4	Test error	

Tab. 7: Pin assignment of the switching outputs in the “Direction-dependent signals” operating mode

7 Commissioning

7.1 Configuration



The AOS104 RTG is configured using the SOPAS configuration software and the Flexi Soft Designer.

LMS - SOPAS

The monitoring fields and evaluation cases required to prevent collisions are already available in the laser scanners.

- Adapt the geometry of the monitoring fields in SOPAS to the existing conditions on-site.



You can find additional information on configuration of the laser scanners and using the SOPAS configuration software in the AOS LiDAR operating instructions.

Flexi Soft

The configuration required to operate the AOS104 RTG system is available in the project file on the USB stick supplied.

- With the aid of the Flexi Soft Designer, transfer the configuration via file download to the Flexi Soft control.



You can find additional information on downloading the configuration files with the Flexi Soft Designer in the AOS LiDAR operating instructions or in the separate **Flexi Soft QuickStart** instructions on the USB stick.

7.2 Checking the stopping and warning response of the system

After aligning the laser scanners and configuring the system with SOPAS and the Flexi Soft Designer, test for the correct stopping and warning response from the application.

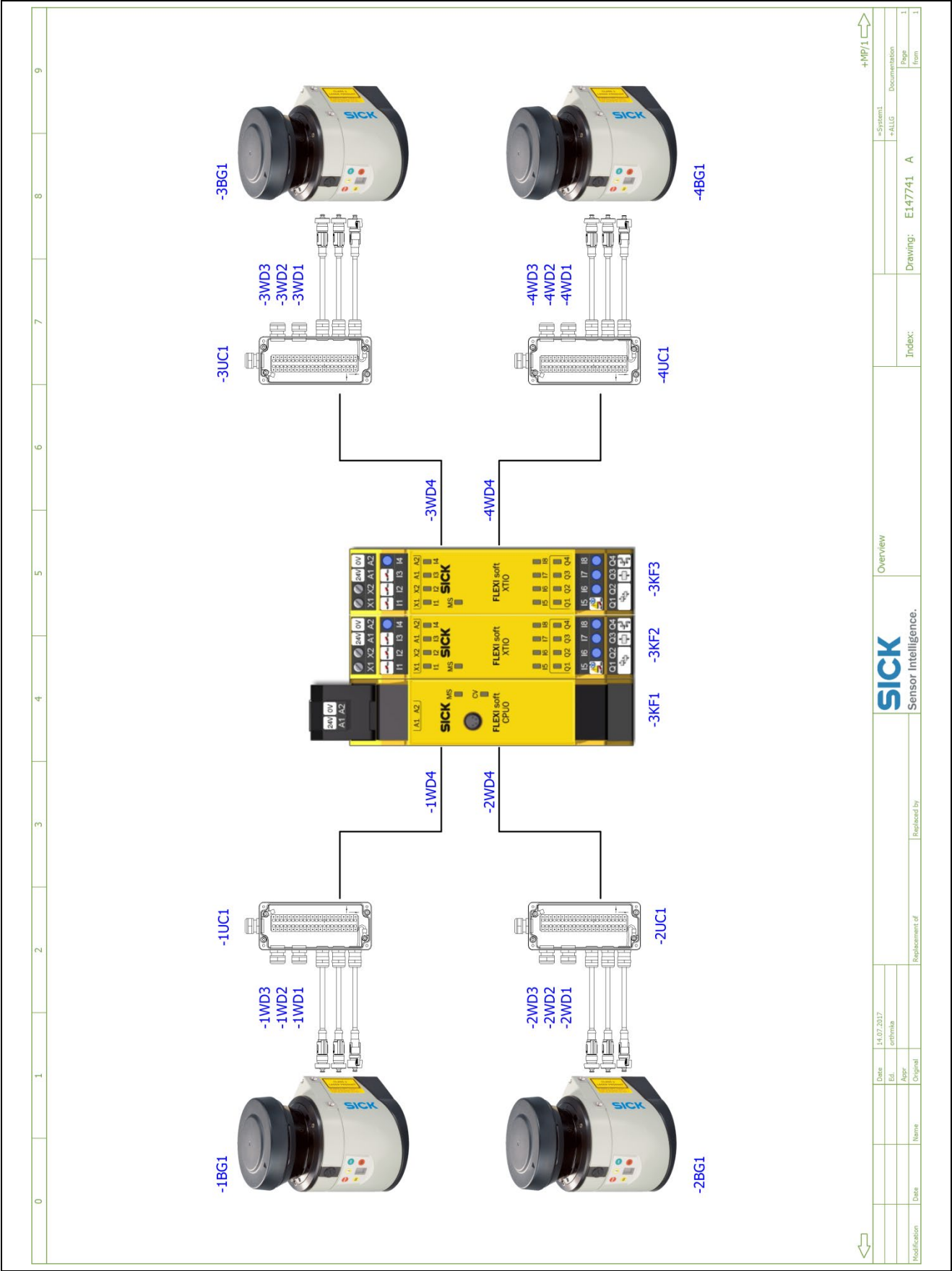
Checking object detection

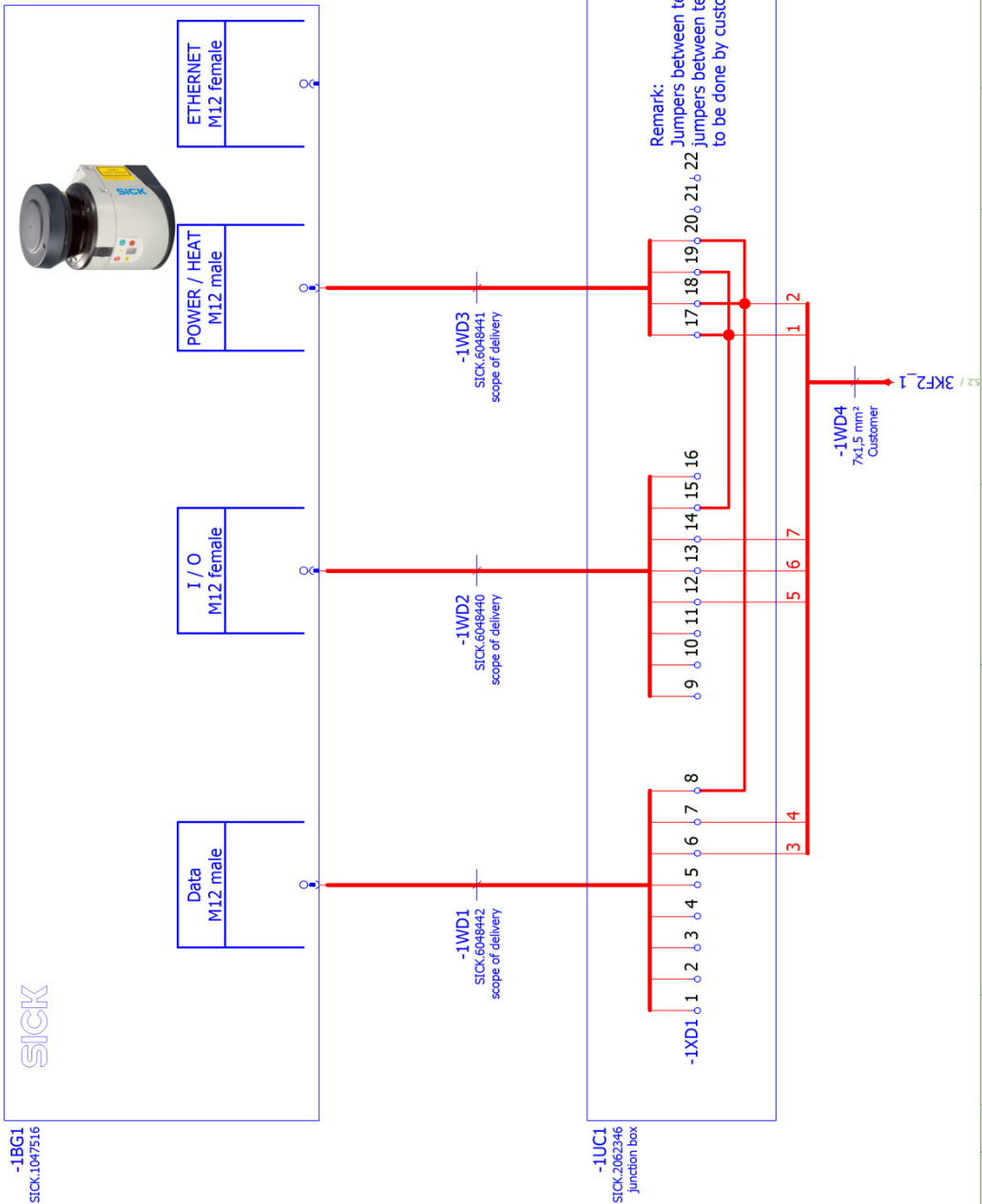
- Check the monitoring fields along their outer edges for proper detection. To do so, walk the edges yourself, or check the functionality using tools suitable for your application. The system should trigger a response accordingly for every field breach.

Ensuring proper functioning within the overall system

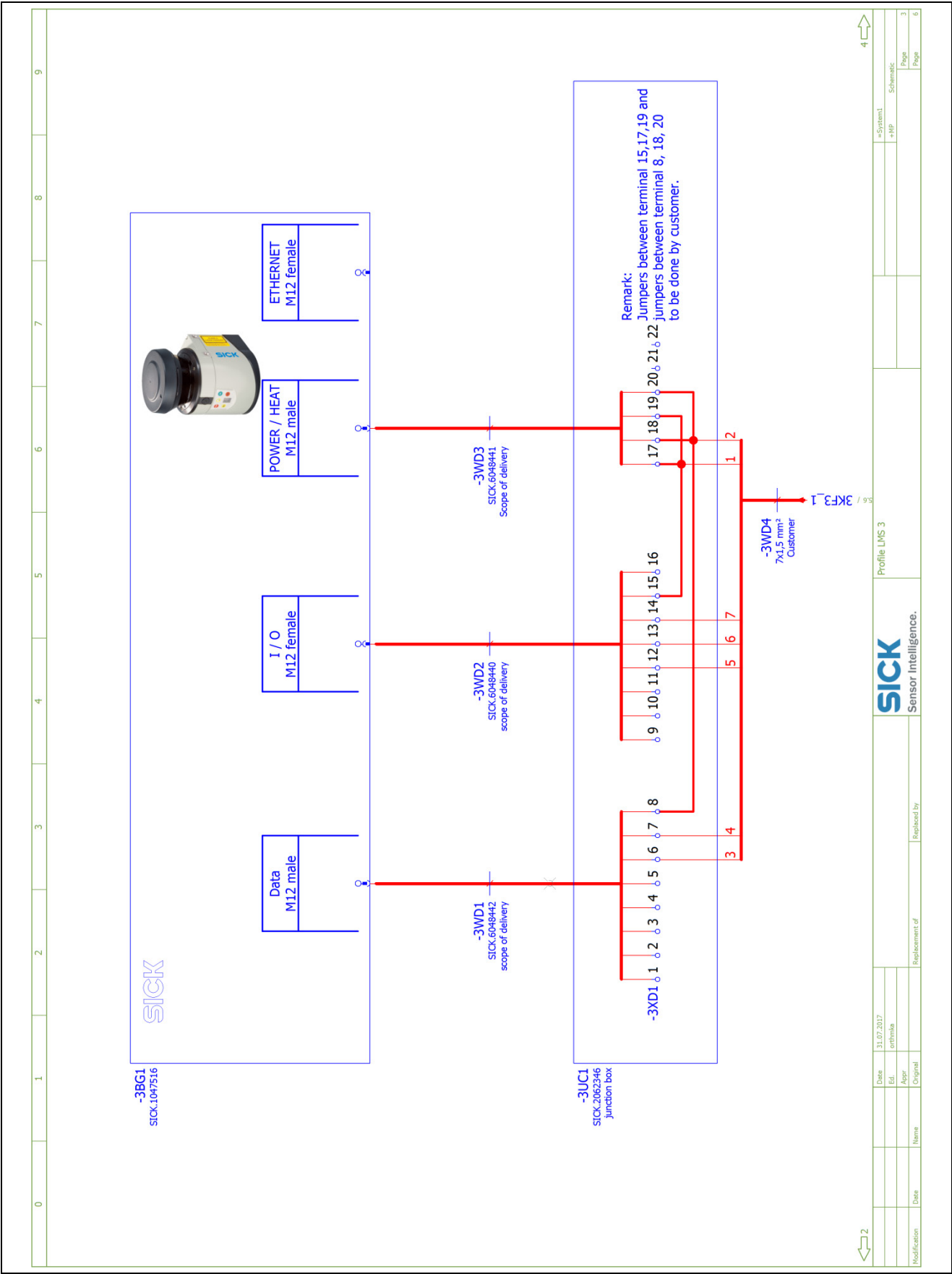
- Whenever system commissioning occurs, check for the required functionality of the AOS104 RTG within the overall system, as well as the interfaces necessary for trouble-free operation.

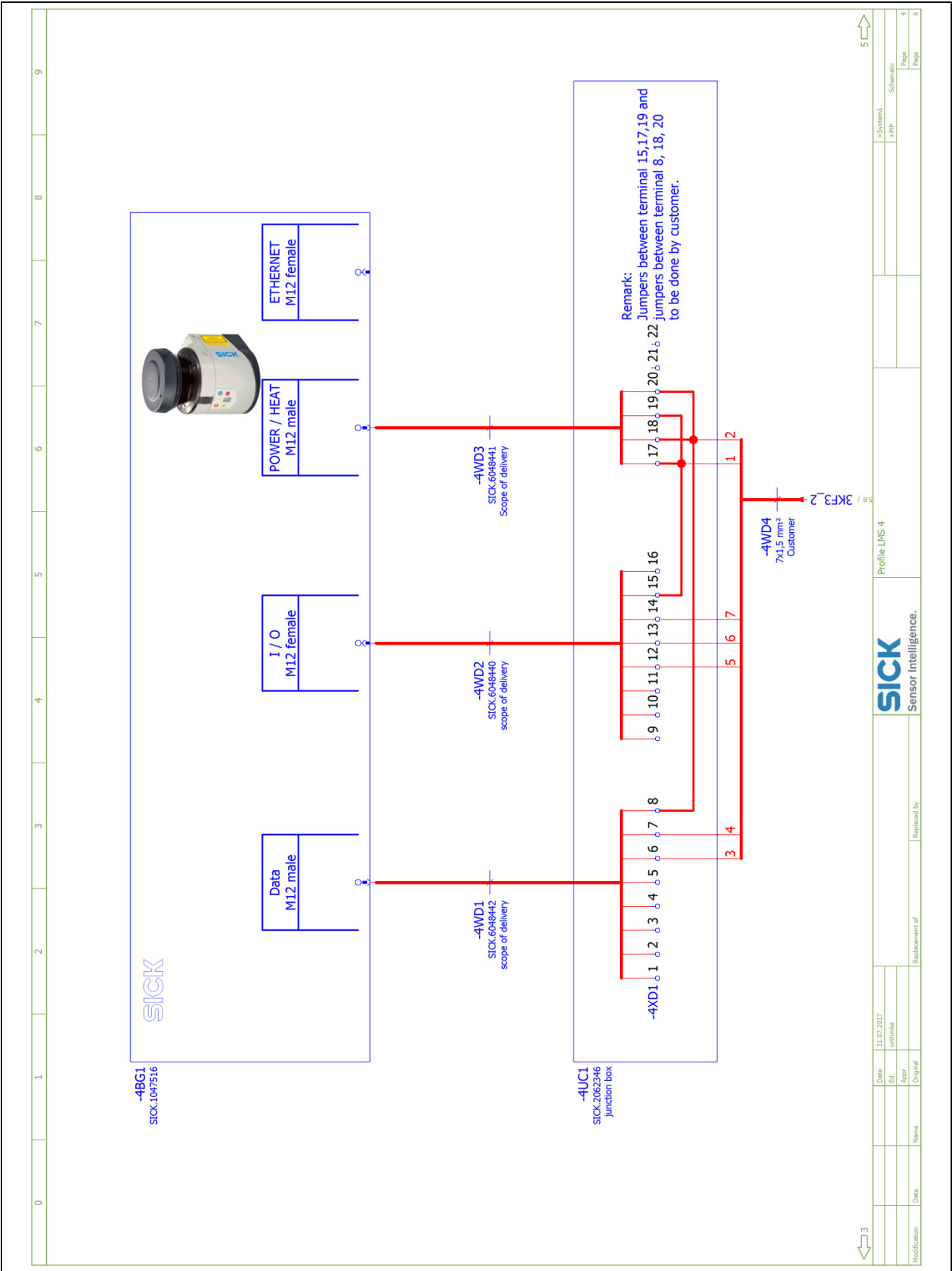
8 Electrical wiring diagrams

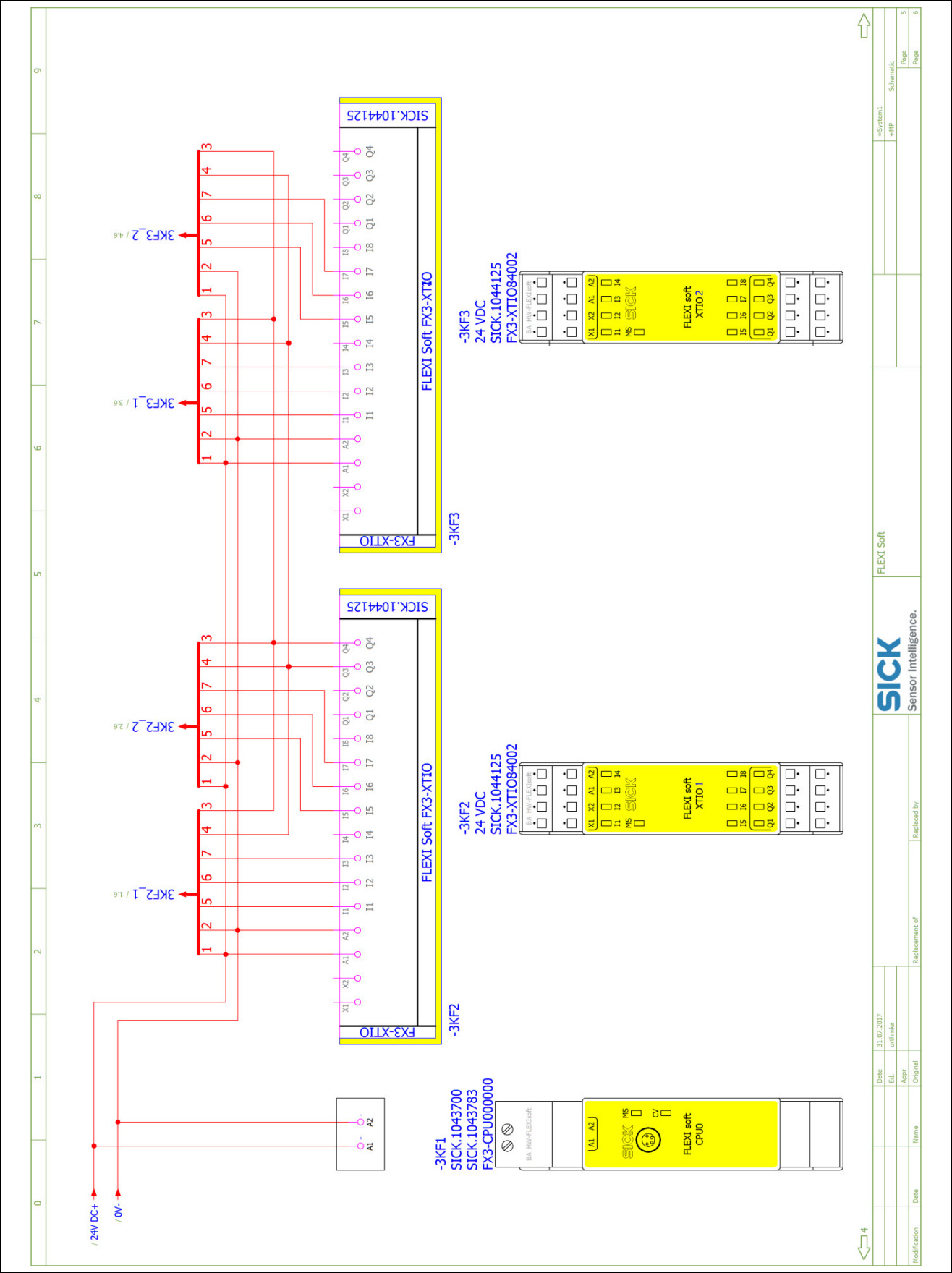












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